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Shiva Sankeerth Reddy Yarradla

Stony Brook, New York

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Education

Master of Science in Computer Science, *SUNY - Stony Brook University* — Expected December 2022

Bachelor of Engineering in Information Technology, *Osmania University ; GPA : 3.9/4.0* — August 2015 — May 2019

Skills

Programming Languages	Python, Java, C++, Javascript, HTML/CSS, SQL, MongoDB
Tools & Libraries	TensorFLow, Pytorch, Keras, Git, Flask, Django, Pandas, Numpy, Dask, Scikit-learn, OpenCV, NLTK, Spark
Cloud Technologies	Docker, Kubernetes, Amazon Web Services, GCP AI Platform, Azure ML
Relevant Courses	Data Structures, Algorithms, Big Data Analytics, Probability and Statistics, Data Science, Database Systems, Natural Language Processing, Artificial Intelligence, Data Mining

Technical Experience

Graduate Researcher Assistant — **December 2021 — Present**
Knowledge Systems Lab, Stony Brook University — *Stony Brook, US*

- Working with Prof. I.V. Ramakrishnan at Knowledge Systems Lab on using deep learning models for classification of intrapartum signals(Fetal Heart Rate and uterine activity) to determine if C-section is required or not.
- Implemented GAN model with one-sided label smoothening to generate fetal heart rate signal images with generator loss of 2.32.
- Implemented Bayesian network methods for FHR pattern classification and developed a novel density estimation technique for nonparametric feature classification and software for rule-based FHR tracing categorization.

Applied Natural Language Processing Engineer — **October 2020 — July 2021**
BeamX TechLabs Pvt. Ltd. — *Hyderabad, IN*

- Designed and developed the pipeline of virtual assistant chat bot capable of engaging in dialogues and question-answering from textual/tabular documents using RASA and BERT-QA and improved the accuracy from 65% to 95% for classifying long documents into multiple classes using hyperparameter tuning.
- Optimised the model's inference time by 34% using techniques like Quantization.
- Built custom training jobs on Google Cloud ML Engine for Object tracking and Object Detection. This helped to reduce the training time by 20% and perform hyper-parameter tuning efficiently in a distributed manner.

Associate Business Technology Analyst — **June 2019 — October 2020**
Deloitte Touché Tohmatsu Ltd. — *Hyderabad, IN*

- Developed and deployed Data collection and pre-processing pipelines as AWS Lambda step functions to pre-process the Loan Notice Document Images to improve extraction accuracy.
- Overhauled the OCR back-end using Tesseract OCR which enhanced the overall extraction efficiency from 87% to 94%.
- Formulated OCR extraction patterns using regular expressions to extract data from Loan Notice Documents.
- Developed a service to automatically perform a set of system tests on extraction patterns bringing down the time to fix issues by 50%.

Publications

- A Novel Approach to Detect Malaria from Cell Images using Deep Learning Approaches. View Publication — September 2020

Projects

Agitation Detection in Persons with Dementia using Machine Learning from Multi-modal data — **December 2021 — Present**

- Working with Prof. Shubham Jain at Mobile Systems Lab on early detection of agitation episodes in dementia patients using machine learning on data acquired from wearable sensors and implementing a Siamese Network model to estimate the efficacy of CBD-oil.

Natural Language Query to SQL Query generation using Content Enhanced BERT — **October 2021 — December 2021**

- Transformed Natural Language Queries from WikiSQL Dataset into SQL queries by integrating database design principles as a vector to the BERT model. SQLNet with bi-directional LSTM was used as a baseline, and it was improved using content enhanced bert. View Project

End to End Speech Recognition for Telugu — **October 2021 — December 2021**

- Converted Input features into MFCC instead of a power-spectrograms, reducing the input size drastically without affecting performance.
- Adapted this system for Low Resource Languages like Telugu too by Scrapping data from Wikipedia to use as a corpus for training language model.
- Integrated Language model with prefix beam decoding to reduce word error rate by 1.7%.
- Built custom training jobs on AWS Sagemaker for efficient hyper-parameter tuning.

Flight delay prediction using a Two-Stage Predictive Model — **July 2021**

- Implemented a Decision Tree Classifier to predict if a flight will miss on-time arrival and combined it with Random Forest Regressor to predict the delay time in minutes. View Project

Leadership Activities & Achievements

- Leading Computer Science Graduate Student Organisation of 500 students as a Senator.
- Among 10 awardees to get Best New Hire recognition at Deloitte FinTech Division.
- Outstanding Project Award from the Department of IT among 50 proposals for Undergraduate Final Year Major Project.
- Secured 1st Prize in IEEE South India Regional Conference Hackathon and Project Expo.